A 100 MHz Charge-Pump PLL with a Current-Starved Ring VCO and an In-Macro TSPC Prescaler on One GF180 Tiny Tapeout Tile

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https://github.com/Hive-RSI-Lab/ece298a-pll

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Abstract

We present a type-II, third-order charge-pump phase-locked loop (PLL) for a single 1×2 Tiny Tapeout tile in the GF180MCU 3.3 V process. A self-biased charge pump with current steering and a rail-to-rail dump-node buffer drives a passive second-order loop filter built from MOS capacitors; a five-stage current-starved ring VCO with a degenerated V-to-I converter, a floor current and a bias-tracking ceiling sets the output frequency. A true-single-phase-clock (TSPC) divide-by-2 inside the analog macro hands the standard-cell logic half the VCO frequency, which lifts the digital half's worst setup slack at the slow corner from about 0 ns to 2.34 ns. The digital half (phase-frequency detector, /4 prescaler, programmable $/N$ divider, lock detector and runaway recovery) is formally checked, fully covered and mutation-tested. After layout and parasitic extraction of the final cell revision the VCO spans 4.2 MHz to 12.9 MHz to 183 MHz to 298 MHz at its ends over 31 process, voltage and temperature (PVT) corners, the loop keeps at least 40.5° of phase margin, and a mixed-signal co-simulation of the extracted analog cell with the hardened gate-level digital netlist and its SDF locks at 100 MHz (6.25 MHz × 16) within 7.2 µs at all six corners that bound the range, with the control voltage settled to 15.0 mV peak-to-peak or better, and relocks to 150 MHz after a divider step within 4.9 µs. The tile passes DRC, LVS and the Tiny Tapeout precheck. All of this is pre-silicon simulation: no chip has been measured. Tests added after an outside review found limits on the previous cell revision that have not been re-run on the final one: with the 5 pF control pad, phase margin fell to 23.9° (the loop still locked); free-running period jitter was 10–14 ps rms across the corners run; and under device mismatch the charge pump's UP/DN bound failed in a quarter of samples at its worst corner.

1 Introduction

Phase-locked loops are the standard way to synthesise an on-chip clock that is a rational multiple of a reference [1, 2]. The charge-pump PLL in particular gives zero static phase error with a passive filter and a simple phase-frequency detector (PFD). This work implements one on Tiny Tapeout's GF180 analog shuttle, which constrains the design in three ways that shape most of the decisions below: the whole PLL, analog and digital, must fit a 347×325 µm tile; Metal5 is reserved for the shuttle, so the process's MIM capacitor (whose top plate is Metal5) is unavailable; and the logic is built from the `gf180mcu_fd_sc_mcu7t5v0` standard cells, whose fastest toggle flop is too slow, at the slow corner, to divide the VCO at its own ceiling.

The design was carried from specification to a precheck-clean GDS with open tools only (ngspice, Magic, Netgen, KLayout, Yosys, OpenROAD/LibreLane, Icarus Verilog, cocotb and cocotbext-ams). Every number in this paper is taken from a result recorded in the repository, and every figure is regenerated from those records by `docs/paper/make_figs.py`.

Section 2 gives the architecture and frequency plan, section 3 the analog circuits, section 4 the digital half, section 5 the loop dynamics, section 6 the physical design, section 7 the post-layout and closed-loop results, and section 8 the limitations.

2 Architecture

Figure 1 shows the loop. The reference on `clk` (5 MHz to 12.5 MHz) and the divided VCO clock `clk_fb` drive a tri-state PFD, whose UP and DN pulses switch the charge pump into the loop filter. The filter voltage v_{ctrl} sets the ring VCO's frequency. The VCO is divided by 2 in the analog macro, by 4 in a ripple prescaler and by a programmable $N \in \{1, \ldots, 5\}$, so

$$f_{\mathrm{vco}} = 8N\, f_{\mathrm{ref}}, \qquad P = 8. \tag{1}$$

The nominal operating point used for every closed-loop test is $f_{\mathrm{ref}} = 6.25$ MHz, $N = 2$ ($PN = 16$), $f_{\mathrm{vco}} = 100$ MHz. Six digital signals cross between the halves (`vco_div2`, `pfd_up`, `pfd_dn`, `pll_en`, `cp_trim[1:0]`); v_{ctrl} and an optional external bias resistor reach the analog pads `ua[0]` and `ua[1]`.

3 Analog design

The macro holds 100 devices in eight blocks: an enable inverter, the bias, the charge pump with its dump-node

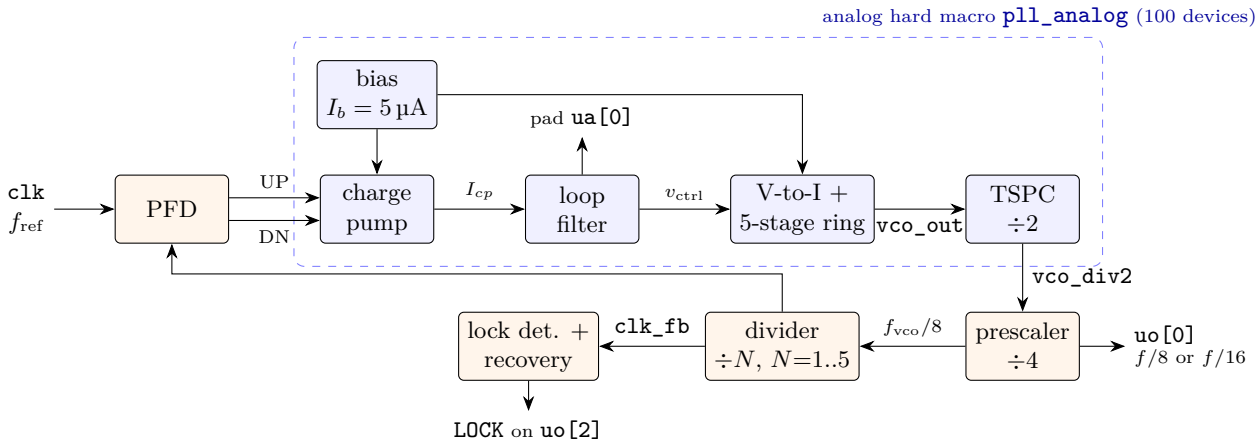


Figure 1: Block diagram. Blue: the transistor-level analog macro. Orange: standard-cell logic. Locked, $f_{\mathrm{vco}} = 8N f_{\mathrm{ref}}$.

buffer, the loop filter, the V-to-I converter, the ring, the output buffer and the TSPC divider. All transistors are 3.3 V `nfet_03v3`/`pfet_03v3`; resistors are unsalicided P+ poly (`ppolyf_u`).

3.1 Bias

A V_{GS}/R self-biased reference sets $I_b = V_{GS,B1}/R_{\mathrm{bias}} \approx 5\,\mu A$ with $R_{\mathrm{bias}} \approx 180\,k\Omega$. Biasing M_{B1} near its zero-temperature-coefficient current density leaves the resistor's tempco as the main temperature dependence. Because the reference's loop also contains a positive-feedback path through the PMOS mirror, a pad capacitance placed directly on the loop node made it relax (a 1.3 µs sawtooth in simulation). Splitting the resistor so that R_{top} isolates the loop node from the pad keeps the input admittance's real part positive for any pad capacitance once $g_{m1} g_{m2} R_{\mathrm{top}}^2 > 1$. A start-up branch releases once the reference conducts; at `ss_m40c_vm10` it needed $L = 4\,\mu m$ on the release transistor to avoid a second operating point.

3.2 Charge pump

The pump current is a trimmed mirror of I_b,

$$I_{cp} = (4 + t_0 + 2t_1 + t_0 t_1)\, I_b = 20/25/30/40\,\mu A, \quad (2)$$

shared by the UP source and DN sink. Long-channel ($L = 2\,\mu m$) source and sink devices keep UP/DN mismatch low over $v_{\mathrm{ctrl}} \in [0.3, V_{DD} - 0.3]$. Current steering keeps both sources always on: when idle their current flows into a dump node that a complementary-input, rail-to-rail unity-gain OTA holds at v_{ctrl}, so a switch event finds its source node already at the output voltage and injects little charge. Without the buffer, the floating dump node gave 25 fC of net charge per event against a 10 fC bound; with it the worst case over 31 corners is 3.6 fC (fig. 2b).

3.3 Loop filter

The filter is the classic series $R_1 C_1$ with a shunt C_2:

$$Z(s) = \frac{1 + sR_1C_1}{s(C_1 + C_2)\left(1 + sR_1 \frac{C_1 C_2}{C_1 + C_2}\right)}, \quad (3)$$

with $R_1 \approx 27\,k\Omega$ (a 150 µm poly resistor), $C_1 = 20.1\,pF$ and $C_2 = 2.1\,pF$. These place the zero at 293 kHz and the pole at 3.06 MHz. Since Tiny Tapeout's GF180 stack forbids Metal5, both capacitors are `cap_nmos_03v3_b` accumulation-mode MOS capacitors (about $3.98\,fF/\mu m^2$ at 1.6 V) in place of MIM: C_1 is twelve $35 \times 12\,\mu m$ units. The MOS capacitor loses about 9 % of its value near $v_{\mathrm{ctrl}} = 0.3\,V$, which only affects the lowest output frequencies. Moving from MIM to MOS shifted phase margin by +0.1 to +0.3° and the post-layout VCO by at most about 1 %.

3.4 VCO

The control voltage drives a source-degenerated V-to-I converter,

$$I_{\mathrm{ctl}} = I_{\mathrm{floor}} + \frac{v_{\mathrm{ctrl}} - V_{TH} - V_S}{R_{\mathrm{V2I}}}, \quad (4)$$

whose current is mirrored into the PMOS and NMOS starving rails of a five-stage inverter ring, giving $f \approx I_{\mathrm{ctl}}/(5\,C_{\mathrm{node}} V_{DD})$. Two additions bound the range. A floor current (a copy of I_b) keeps the ring oscillating at $v_{\mathrm{ctrl}} = 0.3\,V$. A ceiling device in series with R_{V2I}, also mirrored from I_b, saturates at high v_{ctrl} and caps f_{max}; without it the ring reached 390–400 MHz at the fast, cold, high-supply corner. A NAND with `pll_en` and an inverter buffer the ring to full swing with 0.03 ns to 0.07 ns edges and a 53 % to 55 % duty cycle. Figure 3 shows the extracted VCO's tuning curves.

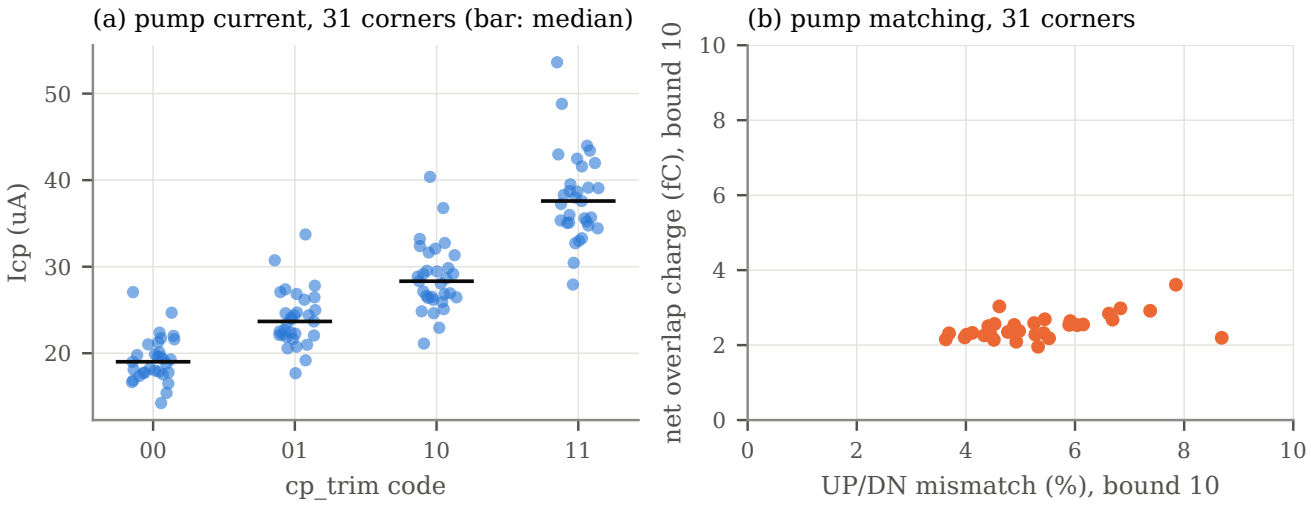


Figure 2: Charge pump over the 31 PVT corners (schematic). (a) UP current per trim code; the code-11/code-00 ratio is 1.96–1.98 at every corner. (b) UP/DN mismatch at code 00 against net overlap charge, both within their 10 % and 10 fC bounds.

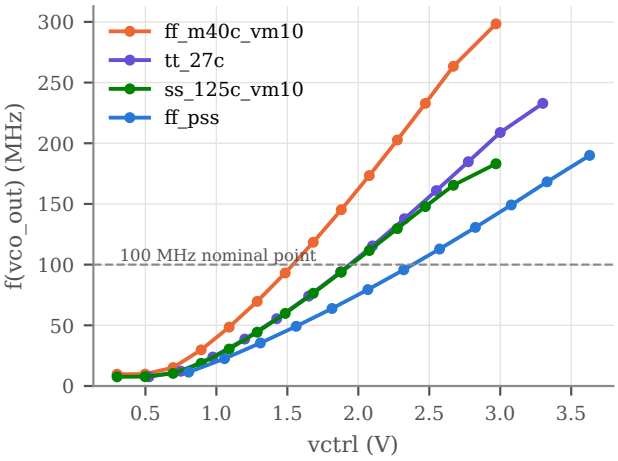


Figure 3: Post-layout VCO tuning curves (extracted netlist, v_{ctrl} forced) at the corners with the highest (`ff_m40c_vm10`) and lowest (`ss_125c_vm10`, 183 MHz at $v_{\mathrm{ctrl}} = V_{DD}$) ceiling, at typical, and at `ff_pss`, the corner with the least pump and V-to-I bias. The flattening at the top is the bias-tracking ceiling. `ff_pss` reaches 100 MHz at $v_{\mathrm{ctrl}} \approx 2.4$ V (interpolated between bench points), where the closed loop settles at that corner (table 2). Points below about 5 MHz are under the bench's resolution and omitted.

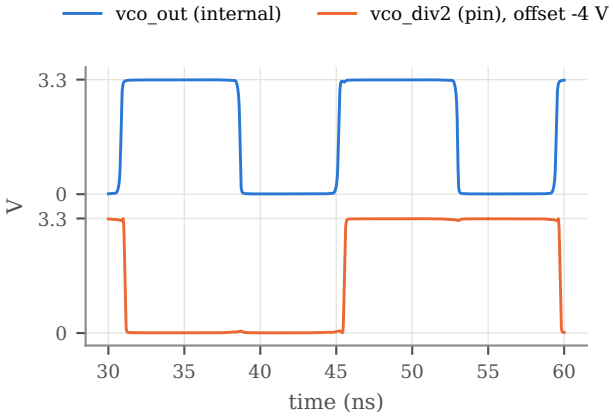


Figure 4: The TSPC divider on the extracted netlist (`tt_27c`, $v_{\mathrm{ctrl}} = 1.6$ V): `vco_out` and the divided `vco_div2`, which toggles on every rising edge.

3.5 TSPC divide-by-2

The first divider stage must toggle at whatever the VCO reaches, including during acquisition overshoot; if it drops edges, `clk_fb` looks slow, the PFD pumps UP, and the loop runs away to the rail. At the slow corner no library flop does this with margin: the best `dffrnq_2+inv_1` toggle loop closed at −0.03 to +0.004 ns of setup slack depending on routing. We therefore placed a nine-transistor Yuan–Svensson TSPC flop [3] with $D = \overline{Q}$ inside the analog macro. Three dynamic stages (precharge on clock low, evaluate on the rising edge) toggle the output node on each rising edge of `vco_out`; a weak PMOS gated by `pll_en` holds the dynamic node high in standby so the cell is fully static there, and two inverters buffer the result to the `vco_div2` pin. All 14 devices use $L = 0.28$ µm. Figure 4 shows its output at the typical corner.

4 Digital design

The digital half is synthesised RTL, 112 standard cells before placement.

- **PFD**: two reset flops clocked by `clk` and `clk_fb`; an

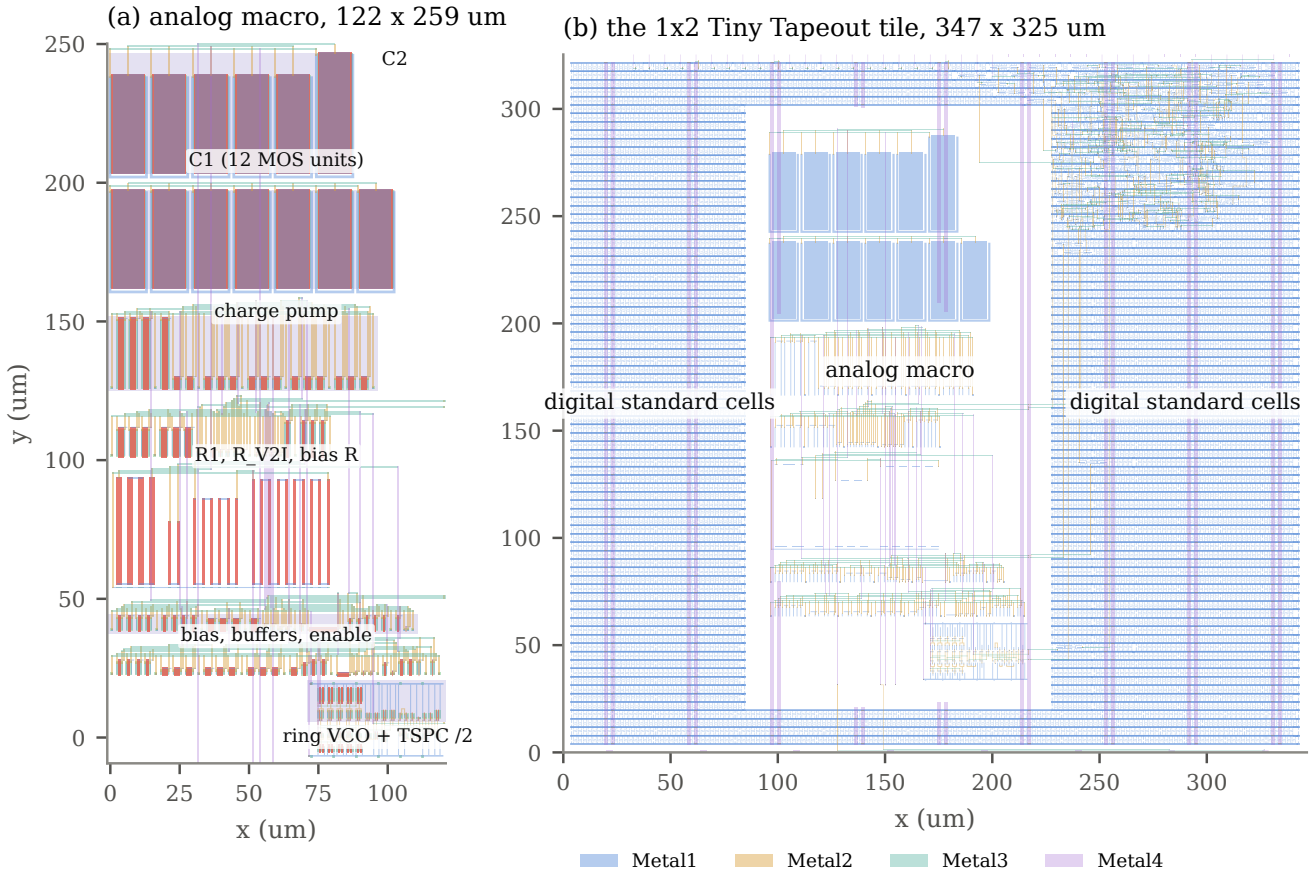


Figure 5: (a) The analog macro as drawn by its generator. (b) The full tile: the analog macro in the middle, the standard-cell digital half on both sides, and the Metal4 power straps.

AND of both outputs resets them through a delay, so both pump switches are on together for a short pulse every cycle, which removes the dead zone.

- **Prescaler**: two ripple toggle flops divide `vco_div2` by 4 (VCO/8); a third gives VCO/16 for observation.
- **Divider**: $N = 1..5$ from `ui[2:0]`; unsupported codes decode as $N = 1$ so the divider never stalls.
- **Lock detector**: a UP or DN pulse still high at the falling edge of the reference is "wide"; `LOCK` rises after 16 consecutive reference cycles without one and falls on the next.
- **Runaway recovery**: if the loop is enabled and unlocked for 1024 reference cycles, DN is forced for 128 cycles, pulling v_{ctrl} to ground. The window (82 µs at 12.5 MHz) exceeds the 50 µs lock-time bound, so it never interrupts a normal acquisition.

Verification: line and toggle coverage are 100 %; six safety properties (divider ratio, prescaler ratio, PFD reset and no-overlap, lock behaviour, asynchronous reset) are proven unbounded, with eight cover points reached; all 16 non-equivalent RTL mutants are killed by the testbench (3 more are equivalent); and gate-level simulation of the hardened netlist passes 18 of 18 tests both without and with SDF delays.

5 Loop dynamics

With the PFD and pump modelled in the continuous-time approximation, the open-loop gain is

$$H(s) = \frac{I_{cp}}{2\pi}\,\frac{K_{\mathrm{vco}}}{s}\,\frac{Z(s)}{PN}, \tag{5}$$

and the second-order approximation gives $\omega_n = \sqrt{I_{cp}K_{\mathrm{vco}}/(PN\,C_1)}$ and $\zeta = \frac{R_1}{2}\sqrt{I_{cp}K_{\mathrm{vco}}C_1/(PN)}$. With the values calibrated from the transistor-level cell at the 100 MHz point before the final V-to-I resize ($I_{cp} = 18.8$ µA, local $K_{\mathrm{vco}} = 116$ MHz/V), the loop at $PN = 16$ crosses over at 585 kHz, about $f_{\mathrm{ref}}/10.7$, safely below the sampling limit [1], with 52.6° of phase margin, $f_n = 416$ kHz and $\zeta = 0.71$ (fig. 7). Phase margin falls with PN (55° at 8, 40° at 40) because the crossover moves down toward the zero. This calibration has not been redone on the final cell; its extracted `tt_27c` values are a mean K_{vco} of 74.6 MHz/V and 47.0° of phase margin at $PN = 16$ (table 1), which the closed-loop runs of

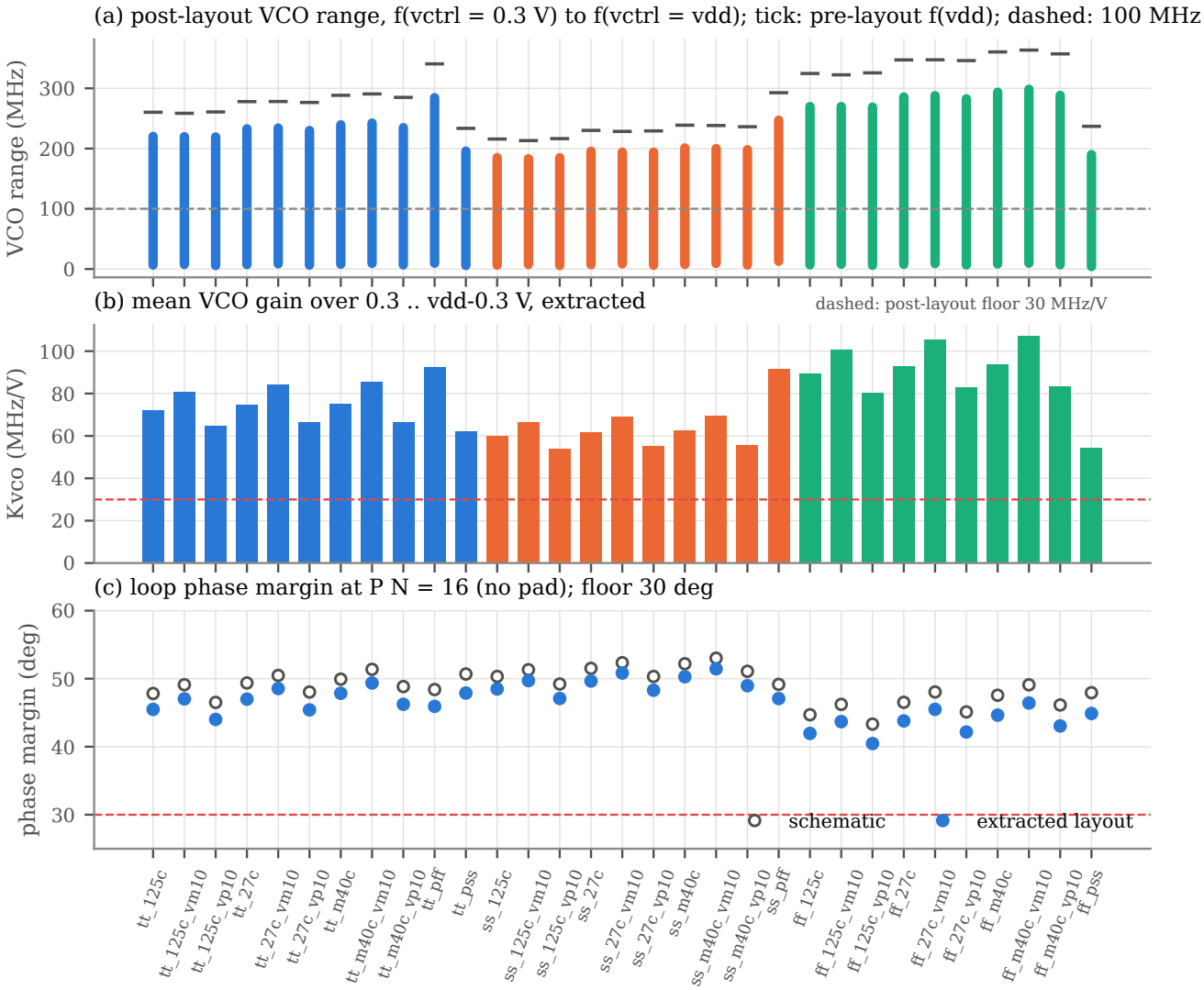


Figure 6: Post-layout results at all 31 corners, coloured by transistor process (blue tt, orange ss, green ff). (a) VCO range from $v_{\mathrm{ctrl}} = 0.3\,\mathrm{V}$ to V_{DD}; ticks mark the schematic's ceiling. (b) Mean VCO gain. (c) Phase margin at $PN = 16$ from each corner's extracted I_{cp}, K_{vco} and filter impedance.

section 7 use directly. A 5 pF pad on v_{ctrl} adds to C_2 and cost 10° to 16° on the previous extracted cell; section 7 reports the loop with it.

6 Physical design

The analog layout is written as code (a gdsfactory/KLayout generator, `gen_pll_analog.py`); nobody edits the GDS. Bottom to top (fig. 5a): the ring VCO with the TSPC beside its output inverter, the bias, buffer and enable rows, the poly resistor serpentines, the pump with its PMOS mirror bank, and the MOS-capacitor rows. The macro is $122 \times 259\,\mu\mathrm{m}$; every routed internal net is labelled so Magic's extraction keeps its coupling capacitance. It is DRC-clean (Magic and KLayout) and LVS-matches the schematic (Netgen, unique match).

The digital half is placed and routed around the macro by LibreLane into the 1×2 tile (fig. 5b; 2348 instances including fill and taps). The VCO-derived clock is constrained per process at twice the extracted VCO's ceiling period plus margin (7.5 ns at ss, 5.4 ns at tt and ff) with 0.20 ns of jitter-derived setup uncertainty. Every STA corner meets setup and hold with no waiver (fig. 8); before the TSPC, the slow corner's first stage closed within a few picoseconds at best. The tile passes full-chip DRC (Magic 0, KLayout 0), LVS and all 12 Tiny Tapeout precheck checks.

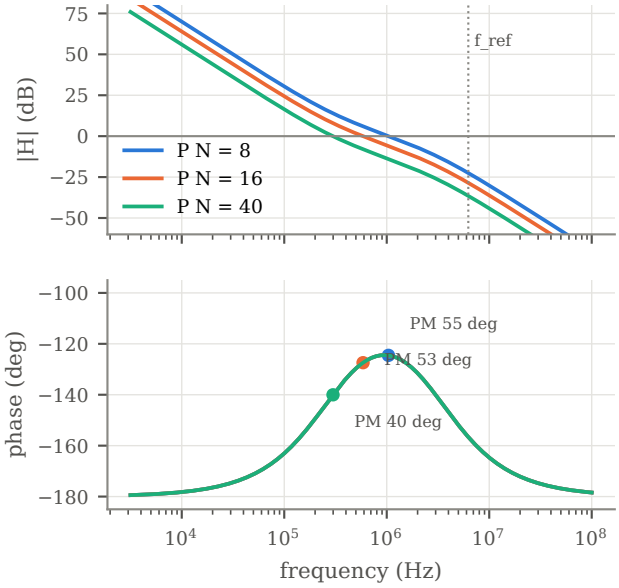


Figure 7: Open-loop gain and phase from the linear model, calibrated at `tt_27c` and 100 MHz on the cell before the final V-to-I resize, for three PN. Dots: crossover.

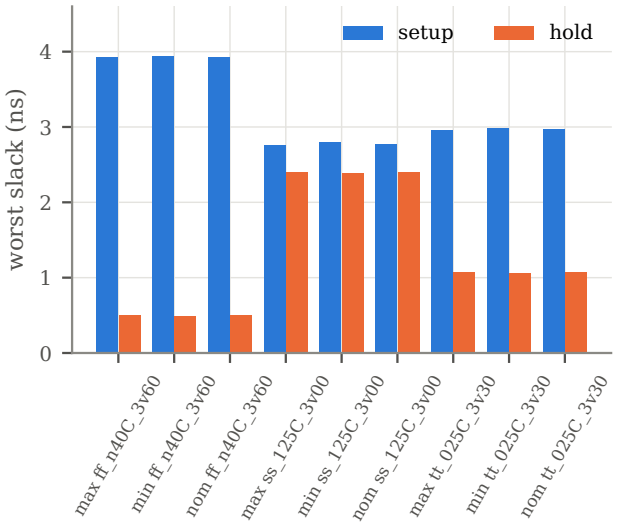


Figure 8: Worst setup and hold slack of the hardened digital half at the nine STA corners (min/nom/max parasitics × ss/tt/ff libraries).

Table 1: Summary over the 31 PVT corners (post-layout unless marked).

Quantity	min	typ.	max
I_{cp}, code 00 (µA)	13.9	18.6	26.6
f_{vco} at $v_{\mathrm{ctrl}} = 0.3$ V (MHz)	4.2	7.4	12.9
f_{vco} at $v_{\mathrm{ctrl}} = V_{DD}$ (MHz)	183	233	298
mean K_{vco} (MHz/V)	53.8	74.6	107.0
phase margin, $PN = 16$ (deg)	40.5	47.0	51.5
UP/DN mismatch (%), sch.	3.6	5.0	8.7
overlap charge (fC), sch.	2.0	2.4	3.6
standby (µA), sch.	1.2×10^{-4}	1.5×10^{-4}	0.026
start-up (µs), sch.	0.43	0.66	0.94

7 Results

7.1 PVT behaviour before and after layout

The analog benches run at 31 corners: tt, ss and ff transistors at −40, 27 and 125 °C and $V_{DD} = 3.3\,\mathrm{V} \pm 10\,\%$, plus four resistor-skew corners (`ss_pff`, `ff_pss`, `tt_pss`, `tt_pff`) where transistors and passives move in opposite directions. Figure 6 summarises the extracted netlist; table 1 gives the ranges.

Layout lowers the VCO's ceiling by 14–20 % (the ring nodes pick up wiring capacitance) and the phase margin by 1.5° to 3.1°. `ff_pss` (fast transistors with slow resistors) still has the least bias, $I_{cp} = 13.9$ µA, but the final V-to-I resize (R_{V2I} length 35→43 µm, a larger clamp and ceiling device) lifted its ceiling from 125.6 MHz on the previous revision to 168.2 MHz, with $K_{\mathrm{vco}} = 54.2$ MHz/V and 44.9° of margin. The lowest ceiling is now at `ss_125c_vm10` (183 MHz at $v_{\mathrm{ctrl}} = V_{DD}$), the lowest K_{vco} at `ss_125c_vp10` (53.8 MHz/V), and the lowest phase margin, 40.5° at `ff_125c_vp10`, clears the 30° floor. Every corner reaches 150 MHz.

7.2 Closed loop on the extracted netlist

The decisive test runs the whole loop in a mixed-signal co-simulation: ngspice simulates the extracted analog macro at a PDK corner, temperature and supply, while Icarus Verilog simulates the hardened gate-level digital netlist with an SDF of the same process at the nearest characterised temperature, the two coupled by cocotbext-ams at a 0.25 ns synchronisation grid. The SDFs for 125 °C ff and −40 °C ss, which the flow does not write, were generated from the same netlist, constraints and extracted parasitics; the same script reproduces a flow-written SDF to within 1 ps. Two mismatches remain: the 27 °C `ff_pss` uses −40 °C delays, the nearest available, and the libraries tie voltage to process (ff 3.6 V, ss 3.0 V), so the digital delays do not follow the analog corner's supply. The corners are the six that bound the post-layout sweep: the lowest and highest VCO ceiling (`ss_125c_vm10`, `ff_m40c_vm10`), the lowest and highest phase margin (`ff_125c_vp10`, `ss_m40c_vm10`), the lowest K_{vco} (`ss_125c_vp10`) and the lowest pump current (`ff_pss`). Each corner locks at $N = 2$ for 30 µs, then the divider steps to $N = 3$ (150 MHz) for 20 µs; a corner takes 38–49 minutes.

Figure 9 and table 2 show the result. The loop locks within 7.2 µs at every corner, `LOCK` never drops after it rises, and the VCO settles to within about 2×10^{-5} of 100 MHz. After the divider step to 150 MHz every corner re-locks within 4.9 µs. `ff_pss`, with the lowest pump

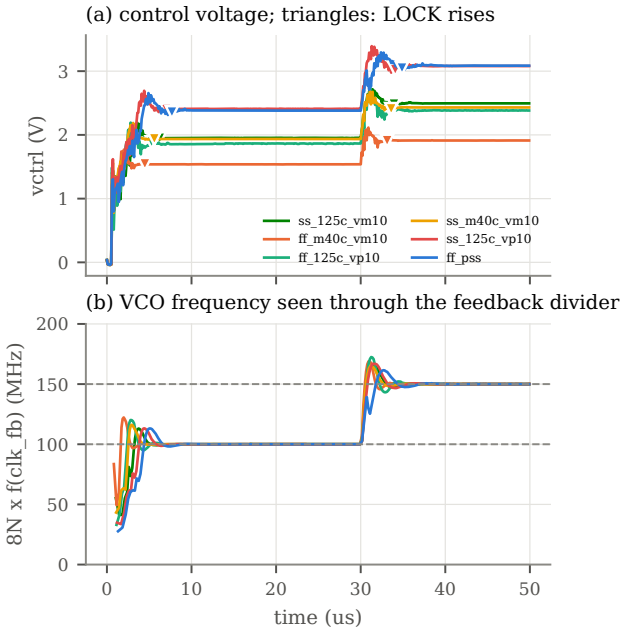


Figure 9: Lock acquisition at the six bounding corners (extracted analog, gate-level digital with SDF), with the divider stepped from $N = 2$ to $N = 3$ at 30 µs: (a) v_{ctrl} with the instants `LOCK` rises; (b) the VCO frequency inferred from successive `clk_fb` periods times $8N$.

Table 2: Closed loop at 100 MHz (6.25 MHz × 16), extracted analog + gate-level digital with SDF, and re-lock after the step to $N = 3$ (150 MHz). Bounds: lock ≤ 50 µs, frequency error ≤ 10^{-3}, v_{ctrl} p-p ≤ 20 mV. SPO: static phase offset.

Corner	lock (µs)	re-lock (µs)	f_{vco} (MHz)	v_{ctrl} (V)	p-p (mV)	SPO (ns)
`ss_125c_vm10`	5.76	3.93	100.0005	1.95	13.9	3.17
`ff_m40c_vm10`	4.00	3.12	100.0021	1.54	7.2	0.90
`ff_125c_vp10`	4.80	3.60	100.0006	1.86	13.4	1.06
`ss_m40c_vm10`	5.12	3.61	100.0004	1.94	4.6	2.37
`ss_125c_vp10`	6.56	4.09	100.0000	2.41	15.0	3.29
`ff_pss`	7.20	4.88	99.9999	2.38	2.6	0.60

current, is the slowest, as the model predicts; it settles with v_{ctrl} at 2.38 V, with `ss_125c_vp10` (lowest K_{vco}) the highest of the six. The static phase offset is 0.60 ns to 1.06 ns at the ff corners and 2.37 ns to 3.29 ns at the ss corners: the PFD's reset pulse plus pump mismatch. On the previous cell revision, timing the digital half by process alone gave `ff_125c_vp10` 3.5 mV and `ss_m40c_vm10` 15.9 mV of ripple, so the ripple depends on the digital delays and these figures should not be read more finely than that.

7.3 Why the gate-level digital matters

Running the same bench with the RTL in place of the gate-level netlist locks just as well, but leaves 9.7–23.1 mV of peak-to-peak ripple on v_{ctrl} (fig. 10); at `ff_pss` the

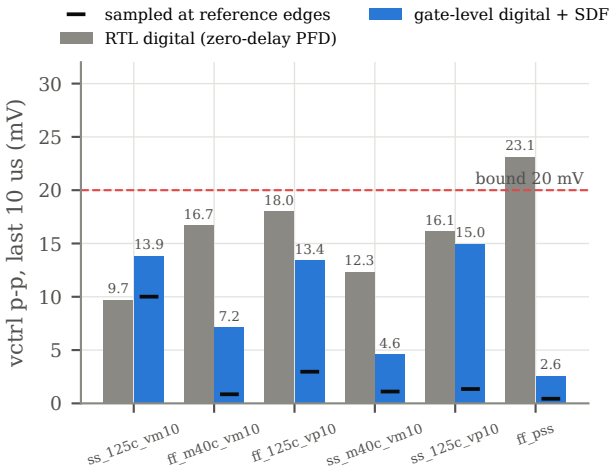


Figure 10: Ripple on v_{ctrl} over the last 10 µs before the divider step, sampled every 5 ns, with the RTL digital and with the gate-level digital. Ticks: the gate-level run's signal sampled at reference edges only.

23.1 mV misses the 20 mV bound, so the RTL run is recorded as failing. The cause is the RTL PFD's zero-delay reset: it never turns both pumps on together, so whatever phase error remains, rounded to the synchronisation grid, is corrected by one pump firing alone, and that lone current through R_1 appears as a spike filtered by C_2. Sampled at reference edges, the RTL's v_{ctrl} moves only 0.01–9.5 mV. The gate-level PFD has a real reset delay, so both pumps overlap briefly every cycle, the correction is a small difference of two currents, and the ripple falls to 2.6–15.0 mV. The lesson is that a zero-delay PFD is fine for checking lock but misleading for ripple and spurs, which depend on the reset pulse.

7.4 With the control-pad load

v_{ctrl} reaches pad `ua[0]`, so the pad's capacitance is on the loop in any real package, probe or no probe. The runs above leave it out. The pad runs below were made on the previous cell revision (before the V-to-I resize) and have not been repeated on the final one. Repeating the then four bounding corners and `tt_27c` with 5 pF from v_{ctrl} to ground inside the extracted cell, all five locked within 7.5 µs (0.2–1.0 µs later than without), held, and met every bound; the pad filters the pump's spikes, leaving 0.8–4.5 mV of ripple (fig. 11). The cost is margin: small-signal phase margin on that extracted cell with the same pad fell to 23.9° at `ff_125c_vp10`, and 9 of the 31 corners were below the 30° floor the no-pad loop is held to. The time-domain runs show this loop stable, not that this margin is comfortable, and 5 pF is the specification's allowance rather than a measured pad.

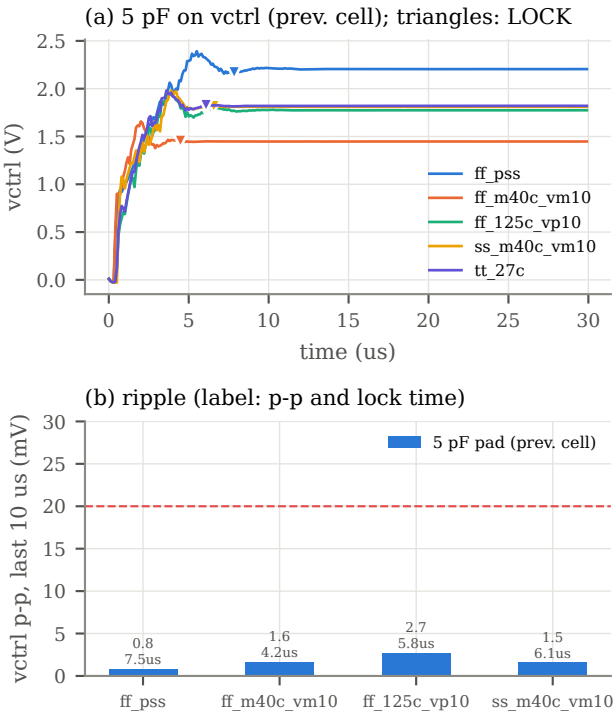


Figure 11: With 5 pF on v_{ctrl} (previous cell revision, extracted analog, gate-level digital): (a) acquisition, (b) ripple and lock time with the pad.

7.5 Other divider settings and re-lock

A second bench changes N while the loop runs, at $f_{\mathrm{ref}} = 6.25$ MHz: $N = 2 \to 3 \to 2$ (100, 150, 100 MHz), 20 µs each, extracted analog with gate-level digital and SDF and no pad, at `ff_pss`, `tt_27c`, `ff_125c_vp10`, `ff_m40c_vm10` and `ss_m40c_vm10` (fig. 12). All five pass: every segment locks or re-locks within 7.2 µs (3.1 µs to 4.9 µs for the steps), and the frequency ends within about 10^{-4} of target (worst 149.9887 MHz at `ff_m40c_vm10`). `ff_pss` reaches 150 MHz with v_{ctrl} at 3.09 V, well below its 168 MHz ceiling. Segment ripple is 2.5–20.2 mV; these segments are not held to the 100 MHz point's 20 mV bound, and at 150 MHz two slightly exceed it (20.2 mV at `ff_m40c_vm10`, 20.0 mV at `tt_27c`).

On the previous cell revision, whose `ff_pss` ceiling was only 125.6 MHz, a step to 120 MHz at $f_{\mathrm{ref}} = 5$ MHz took about 25 µs to settle at that corner, and $N = 4$ (160 MHz) at the lowest-margin corner with the pad rang for tens of microseconds before settling. The resize was made to remove that limit; the 120 MHz step and $N = 4$ have not been repeated on the final cell.

8 Discussion and limitations

What this shows. The evidence is simulation only. It supports "implemented, and demonstrated to lock in post-layout mixed-signal simulation". It does not support

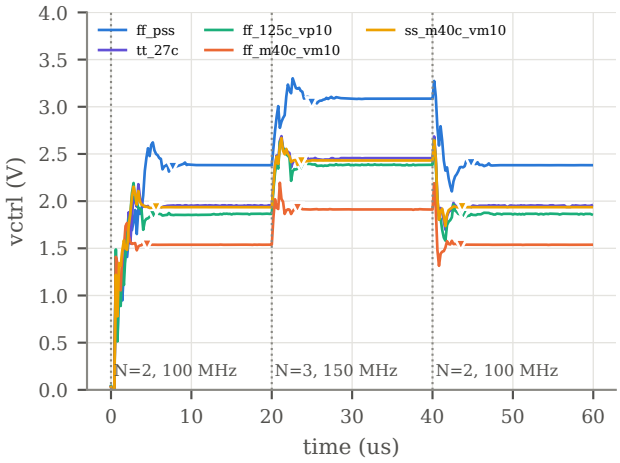


Figure 12: Divider steps $N = 2 \to 3 \to 2$ (100→150→100 MHz) at $f_{\mathrm{ref}} = 6.25$ MHz: v_{ctrl} at five corners (triangles: `LOCK` rises).

"silicon-proven" or "low-jitter".

Lock range. After layout the VCO covers about 13 MHz to 165 MHz at every corner, between the highest bottom (12.9 MHz) and the lowest ceiling (165.5 MHz at the bench's top point, `ss_125c_vm10`); at typical it is about 7 MHz to 233 MHz at $v_{\mathrm{ctrl}} = V_{DD}$. The datasheet promises only the guaranteed range; the slow, hot, low-supply corner now sets the top.

Jitter is estimated, not measured, and the figures below are from the previous cell revision (before the V-to-I resize); they have not been re-run on the final cell. ngspice has no device transient noise, so noise sources sized from each device's model at its bias (white and $1/f$) are injected into the extracted cell, except the output buffer and divider. At the 100 MHz point, over two or three seeds of 520 periods, free-running period jitter is 9.7 ps rms at `tt_27c`, 10.2 ps at `ss_125c_vm10` and 13.5 ps at `ff_125c_vm10`, the highest of the nine corners run (six of them in shorter screening runs; the other 22 were not run). A deterministic alternation of up to ±6.8 ps on the output is separate from these figures. They replace a first estimate of 1.2–2.5 ps, which came from 30–36 periods at the top of the VCO's range with less of the noise included. For the locked output we have only a linear model, not a simulation: from the simulated VCO phase noise and R_1's thermal noise it gives 44–61 ps rms from 10 kHz to 50 MHz at those three corners, or 35–43 ps without the VCO's $1/f$ term, whose level is uncertain by a factor of 2–4 in power. It leaves out reference, PFD, charge-pump and supply noise, so it does not bound the jitter of a real chip. The STA's 0.20 ns setup uncertainty was derived from the first estimate; it still covers 7σ of the worst figure across the two VCO periods of `vco_div2` (about 155 ps), but with far less room, and jitter at the faster clock the STA is constrained to was not simulated. Nothing here measures reference spurs or supply noise.

Device mismatch. The specification had excluded Monte Carlo on the belief that the PDK models carry no per-instance mismatch. They do, for MOS devices. The sampled runs below are from the previous cell revision; the pump they measure was not changed by the V-to-I resize. On the final cell, the analog Monte-Carlo gate (30 seeds on each of seven benches: pump, open-loop, overlap, prescaler, power, start-up and VCO; 210 runs) passes with every run inside its bounds. On the schematic, 200 samples per corner put the worst-case UP/DN pump mismatch over its 10 % bound in 26 % of samples at `ff_pss` (95 % interval 20–33 %; worst 14.6 %) and 6 % at `ff_m40c_vp10`, and in none at `ss_pff` or `tt_27c`. The peak is at the edge of the v_{ctrl} range. So the row in table 1 is the spread with statistics off. At `tt_27c` the overlap charge (worst 3.4 fC, bound 10 fC) and bias start-up hold; the VCO frequency at mid-range is 101.7 MHz with $\sigma = 2.5$ MHz. These are rates given a corner, not a product yield, and they leave out layout gradients, resistor mismatch (the models have none) and capacitor variation; Kvco and phase margin were not run. Closed-loop runs with one pump source widened by about 3σ (extracted cell and gate-level digital at 100 MHz; UP 6 % reads 14.2 % on the same DC sweep at `ff_pss`, UP 5 % reads 12.1 % at `ff_m40c_vp10`) all lock and hold. v_{ctrl} ripple is 1.0 mV to 2.5 mV at `ff_pss` against 2.4 mV unskewed (both on the previous cell), and 3.3 mV to 6.3 mV at `ff_m40c_vp10` against 5.4 mV, inside the 20 mV bound; the static phase offset stays between 0.44 and 0.98 ns. So we keep the 10 % bound as written and do not resize the pump, since at this operating point a pump over that bound does not hurt the loop. This is a widened device, not a drawn sample, at one divider setting; a pump skew during an N step and at the pad-loaded corners was not run.

Coverage of the closed-loop tests. On the final cell the extracted loop has been run at the six bounding corners ($N = 2$, then a step to $N = 3$) and at five corners in the $N = 2 \to 3 \to 2$ bench, not at all 31. The 31-corner sweep characterises analog behaviour; it does not demonstrate acquisition at every corner and setting. $N = 5$ has not been run on the extracted netlist, and $N = 4$ only at one corner, on the previous cell.

Bench strength. A mutation check of the analog benches (each device's width and length doubled or halved) flags mutants that move no measure beyond the run-to-run spread. It now passes: 255 of 295 mutants are killed and the other 40 are ruled to move their measures less than the spread. The TSPC's devices are declared and in the mutated set.

9 Conclusion

A complete charge-pump PLL—bias, pump, MOS-capacitor filter, ring VCO, TSPC prescaler and standard-cell control—fits one 1×2 GF180 Tiny Tapeout tile and is DRC, LVS and precheck clean. Moving the first divide stage into the analog macro as a TSPC flop turned a timing closure measured in picoseconds into 2.3 ns of margin. With parasitics extracted and the real gate-level digital in the loop, the PLL locks to 100 MHz in at most 7.2 µs with at most 15 mV of control ripple at all six corners that bound its range, and re-locks to 150 MHz after a divider step within 4.9 µs. On the previous cell revision it stayed stable with the control pad's load. It has not been fabricated; the phase margin with the pad, jitter and pump mismatch, none yet re-run on the final cell, are the risks to check first.

Reproducing this paper

All results are recorded in the repository with hashes of their inputs. `docs/paper/sims/` re-runs the two post-layout ngspice benches behind figs. 3 and 4; `docs/paper/make_figs.py` redraws every figure; `make -C docs/paper` builds this PDF. `blocks/pll/tb/README.md` explains the co-simulation benches, and `blocks/pll/tb/gl/` holds the gate-level netlist, parasitics and SDFs those runs read.

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